

Problem

A branch voltage in an RLC circuit is described by

$$\frac{d^2v}{dt^2} + 4\frac{dv}{dt} + 8v = 120$$

If the initial conditions are $v(0) = 0 = dv(0)/dt$, find $v(t)$.

Step-by-step solution

Step 1 of 6

Consider the differential equation that describes a branch voltage in an RLC circuit.

$$\frac{dv^2}{dt^2} + 4\frac{dv}{dt} + 8v = 48$$

Apply the Laplace transform to the differential equation.

$$\left[s^2V(s) - sV(0) - V'(0) \right] + 4[sV(s) - V(0)] + 8V(s) = \frac{48}{s}$$

$$V(s)(s^2 + 4s + 8) - V(0)(s + 4) - V'(0) = \frac{48}{s}$$

Substitute 0 for $V(0)$ and 0 for $\frac{dv(0)}{dt}$.

$$V(s)(s^2 + 4s + 8) - 0(s + 4) - (0) = \frac{48}{s}$$

$$V(s)(s^2 + 4s + 8) = \frac{48}{s}$$

$$V(s) = \frac{48}{s(s^2 + 4s + 8)}$$

Step 2 of 6

Decompose $V(s)$ into simple terms using the partial fraction expansion.

$$V(s) = \frac{48}{s(s^2 + 4s + 8)}$$

$$\frac{48}{s(s^2 + 4s + 8)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 4s + 8}$$

$$48 = A(s^2 + 4s + 8) + (Bs + C)s \dots\dots (1)$$

Here A , B , and C are constants.

Step 3 of 6

Substitute 0 for s in equation (1).

$$48 = 8A$$

$$A = 6$$

Equate coefficients of s^2 on both sides of the equation (1).

$$A + B = 0$$

$$B = -A$$

$$B = -6$$

Equate coefficients of s on both sides of the equation (1).

$$4A + C = 0$$

$$4(6) + C = 0$$

$$C = -24$$

Step 4 of 6

Substitute obtained A, B, and C values in equation $V(s)$.

$$\begin{aligned} V(s) &= \frac{6}{s} + \frac{-6s - 24}{s^2 + 4s + 8} \\ &= \frac{6}{s} - \frac{6(s + 4)}{(s + 2)^2 + 2^2} \\ &= \frac{6}{s} - 6 \left[\frac{(s + 2)}{(s + 2)^2 + 2^2} + \frac{2}{(s + 2)^2 + 2^2} \right] \\ V(s) &= \frac{6}{s} - 6 \left[\frac{(s + 2)}{(s + 2)^2 + 2^2} + \frac{2}{(s + 2)^2 + 2^2} \right] \dots\dots (2) \end{aligned}$$

Step 5 of 6

Consider the Laplace transform pairs.

$$\mathcal{L}^{-1}\left(\frac{1}{s}\right) \xleftrightarrow{\mathcal{L}} u(t)$$

$$\mathcal{L}^{-1}\left(\frac{s+\alpha}{(s+\alpha)^2+\beta^2}\right) \leftrightarrow e^{-\alpha t} \cos \beta t u(t)$$

$$\mathcal{L}^{-1}\left(\frac{\beta}{(s+\alpha)^2+\beta^2}\right) \leftrightarrow e^{-\alpha t} \sin \beta t u(t)$$

Determine $v(t)$.

Apply the inverse Laplace transform on both sides of equation (2).

$$\begin{aligned} v(t) &= \mathcal{L}^{-1}[V(s)] \\ &= 6\mathcal{L}^{-1}\left[\frac{1}{s}\right] - 6\mathcal{L}^{-1}\left[\frac{(s+2)}{\left((s+2)^2+2^2\right)} + \frac{2}{\left((s+2)^2+2^2\right)}\right] \\ &= 6\mathcal{L}^{-1}\left[\frac{1}{s}\right] - 6\left[\mathcal{L}^{-1}\left[\frac{(s+2)}{\left((s+2)^2+2^2\right)}\right] + \mathcal{L}^{-1}\left[\frac{2}{\left((s+2)^2+2^2\right)}\right]\right] \\ &= 6[u(t)] - 6\left[\left(e^{-2t} \cos(2t)u(t)\right) + \left(2e^{-2t} \sin(2t)u(t)\right)\right] \\ v(t) &= 6\left[1 - e^{-2t}(\cos 2t + 2 \sin 2t)\right]u(t) \text{ V} \end{aligned}$$

Step 6 of 6

Therefore, the response $v(t)$ of an RLC circuit is $\boxed{6\left[1 - e^{-2t}(\cos 2t + 2 \sin 2t)\right]u(t) \text{ V}}.$